SML Assignment 1

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1. On the `Nine Wheels of SML`

Problem Chosen: Predicting House Prices for a district in Taiwan.

The real estate market plays a key role in any economy, as it reflects the combined effects of demographics, the economy, and location. Knowing the likely price of a house is important, not just for buyers and sellers trying to make fair deals, but also for policymakers and city planners who want to support sustainable and accessible housing. Many factors can affect house prices. For example, the age and condition of a property, how close it is to public transport or shops, the local economy, and the neighborhood itself all make a difference. Houses near transit stations or popular commercial areas usually sell for more, while older houses or those in less convenient locations tend to be cheaper. In Taiwan, fast urban growth and high population density make reliable property valuations even more important. By looking at things like demographics, location, and economic indicators, it is possible to estimate house prices more accurately. These estimates can help everyone, from homebuyers to government officials, make better decisions about real estate.

Building on this, this project aims to develop a system that predicts house prices per unit area for properties in a district in Taiwan. The system will use key features such as the age of the house, proximity to the nearest MRT station, the number of nearby convenience stores, and the geographic coordinates of the property. These features are particularly important because they capture the factors that most strongly influence a property's value: house age reflects its condition and potential maintenance costs, proximity to public transport affects accessibility and convenience, nearby amenities indicate how desired the area is, and geographic location captures broader neighborhood effects that might not be visible from other data alone. By focusing on these features, we can build a model that provides accurate, continuous-valued price estimates to support informed decisions in the housing market.

Wheel 1: Clear Problem Choice and Formulation

We formulate the task as a regression problem, where the goal is to predict

$$Y = \text{house price per unit area} \in [7.6, 117.5] \subset \mathbb{R}.$$

given a feature vector

$$x = (\text{Date, Age, MRT\_distance, Convenience, Latitude, Longitude}) \in \mathbb{R}^6.$$

The output space $\mathscr{Y} = \mathbb{R}$ is continuous, making regression the natural choice. Predictions are expected to lie roughly within the range of observed prices in the dataset, though the model could, in principle, produce values outside this range. This makes a regression formulation the most natural thing to choose.

Wheel 2: Data Collection and Exploration

We use the public dataset from the UCI Machine Learning Repository.
It contains 414 observations and 6 attributes, including transaction date, house age, distance to the nearest MRT station, number of convenience stores, latitude, longitude, and house price per unit area. These

features form our input space, $\mathscr{X} = \mathbb{R}^6$, with dimension 6. The index $\kappa = 69$ indicates our dataset is an overdetermined system thus indicating stability of the regression estimates. The output space, $\mathscr{Y}$ is a subset of the real line $\mathbb{R}$

We will perform the following exploratory data analysis (EDA) steps:

1. Inspect the distribution and range of each feature.
2. Check for missing values and handle them if any.
3. Examine pairwise correlations to identify multicollinearity.
4. Visualize relationships between features and the target variable using scatterplots and boxplots.

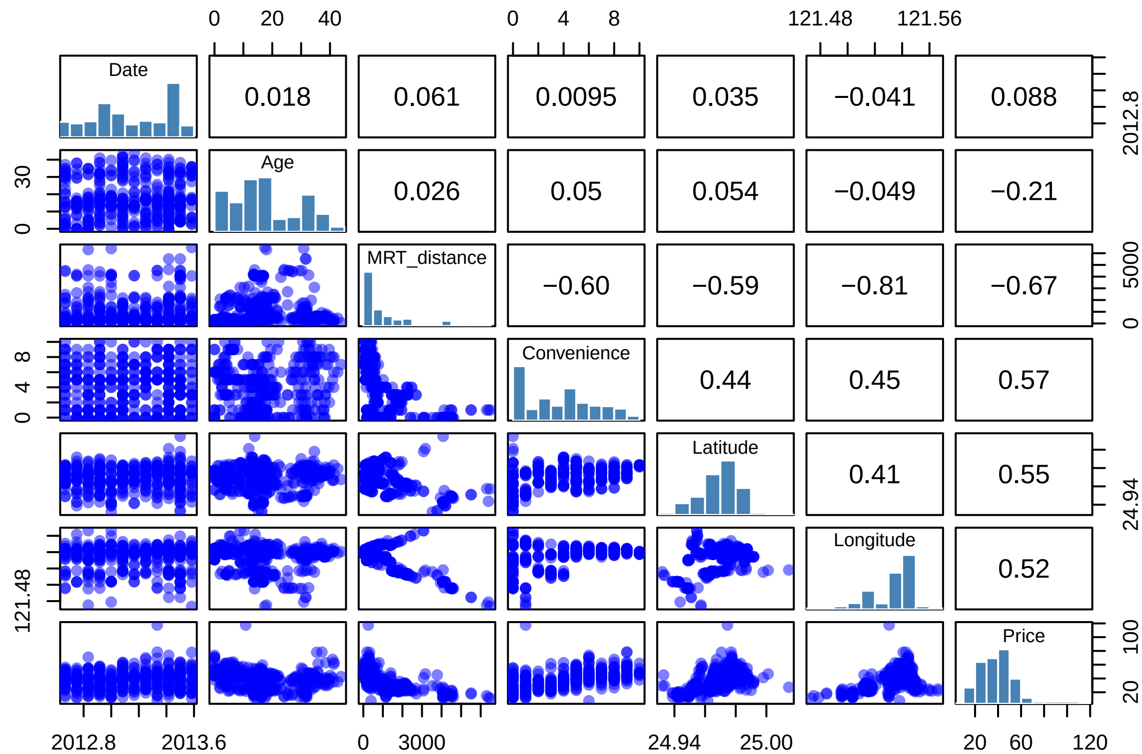
The UCI repository is useful in the situation since it contains historical data on house prices, readily available with all the necessary features. Exploring the data is essential for understanding patterns, identifying outliers, and guiding model selection. Correlation analysis helps detect multicollinearity, which can affect model stability in linear regression. Proper EDA ensures that our model is grounded in the structure of the data and that any preprocessing or feature engineering decisions are well-informed.

Table 1: Summary statistics for Real Estate dataset

	vars	n	mean	sd	median	min	max
Date	1	414	2013.148953	0.2819953	2013.1667	2012.66667	2013.58333
Age	2	414	17.712560	11.3924845	16.1000	0.00000	43.80000
MRT_distance	3	414	1083.885689	1262.1095954	492.2313	23.38284	6488.02100
Convenience	4	414	4.094203	2.9455618	4.0000	0.00000	10.00000
Latitude	5	414	24.969030	0.0124102	24.9711	24.93207	25.01459
Longitude	6	414	121.533361	0.0153472	121.5386	121.47353	121.56627
Price	7	414	37.980193	13.6064877	38.4500	7.60000	117.50000

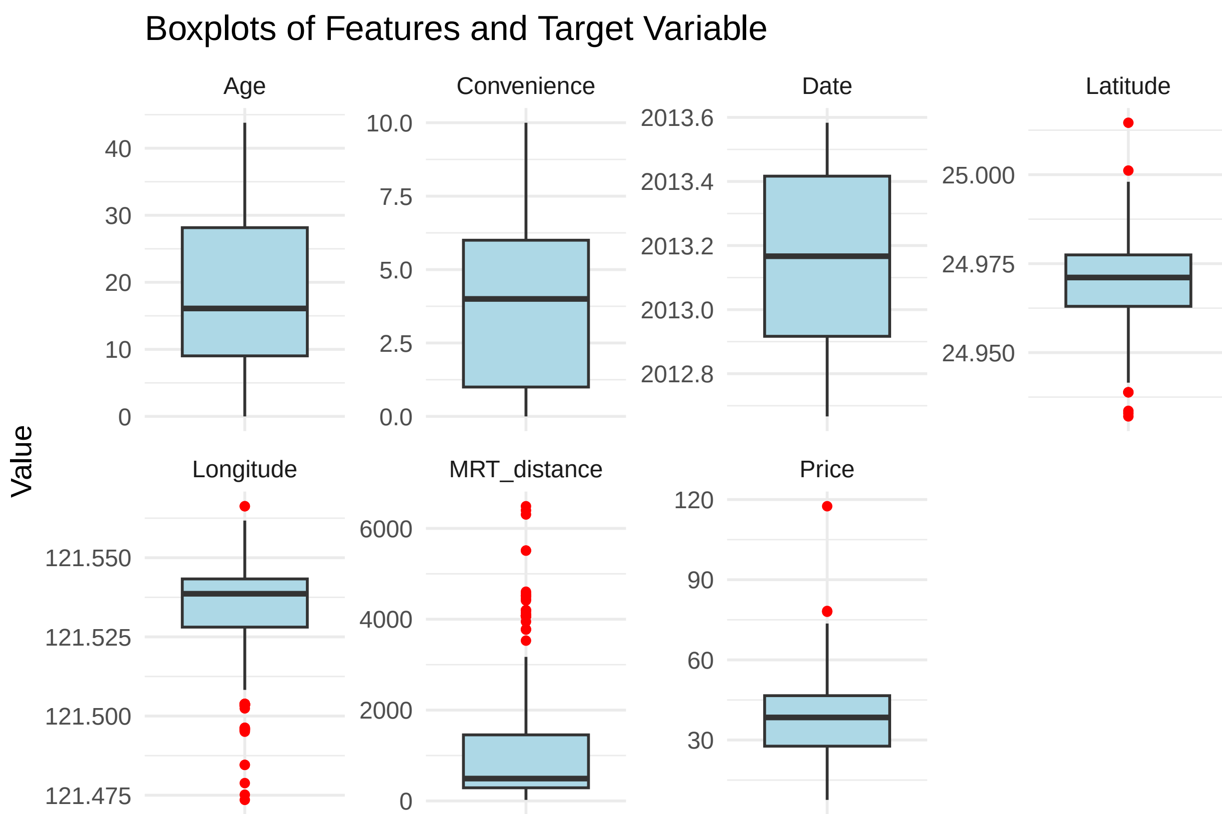
Brief intepretation: The dataset contains 414 observations per variable (no missing values) with six input features and the target variable, `Price`. The features are type homogeneous but scale heterogeneous. For example, the distance to the nearest MRT station ranges from about 23 to over 6,400 meters, while the number of nearby convenience stores goes from 0 to 10. Even though latitude and longitude change only slightly, they can still affect property prices. Comparing the mean and median values shows some skewness in certain features, especially MRT distance and house price, suggesting the presence of outliers or extreme values.

For the regression task, this means a few things. The features may need to be scaled so that differences in units don't skew the model. Handling outliers could improve stability. Distance and location features might also benefit from transformations, such as taking the logarithm, to better capture their impact on house prices.



Brief intepretation: The correlation analysis shows that some features have stronger relationships with house `Price` than others. In particular, `MRT_distance` is negatively correlated with price (around -0.67), meaning houses closer to MRT stations tend to be more expensive. `Convenience`, `Latitude`, and `Longitude` show moderate positive correlations with price, suggesting that location and nearby amenities contribute to higher values. Other features, like `Date` and `Age`, show weaker correlations, indicating a smaller direct effect on `Price`.

The scatterplots reveal some non-linear patterns, especially for `MRT_distance` and `Convenience`, and potential outliers are visible in several variables. This suggests that some feature engineering could improve model performance. For instance, log-transforming distances, creating interaction terms between location features, or considering powers of some variables (like `Age`).



Brief Interpretation:

The boxplots support what we already noticed from the correlation analysis. Location and accessibility features don't vary much. In contrast, features like Age and MRT distance spread out more and show clear outliers. Price is also right-skewed, with a few properties selling for much more than the rest. Overall, the distributions suggest that scaling the features and being mindful of outliers will help the model, but the dataset remains clean and very suitable for regression.

Wheel 3: Hypothesis Space

Based on the analysis from the previous wheel, I restrict my hypothesis space to three classes of regression functions for predicting house prices.

1. Linear Regression Class $\mathscr{H}_{\text{lin}}$

$$\mathscr{H}_{\text{lin}} = \{\, f(X) = X^\top \beta \,\}.$$

This space corresponds to classical linear regression. It is high-bias but low-variance. Its coefficients give information about how each feature affects `Price`.

2. Polynomially Enriched Class $\mathscr{H}_{\text{poly}}$

$$\mathscr{H}_{\text{poly}} = \mathscr{H}_{\text{lin}} \,\cup\, \{\, \gamma_1 \,\text{Age}^2 \}.$$

2. Lograithmic Class $\mathscr{H}_{\text{poly}}$

$$\mathscr{H}_{\text{log}} = \mathscr{H}_{\text{poly}} \cup \{\gamma_2 \log \text{MRT\_distance}\}.$$

This choice of functions is based on the patterns observed in the scatter plots. Some curvature is obvious in the plots of `Price` against each of `Age` and `MRT_distance` even though some parts of the relationship with `Age` looks linear.

Wheel 4: Principle of Learning and Criteria

Since the chosen hypothesis space suggests that the model is linear in the paramters, then Ordinary Least Squares is still a valid principle of learning. Since we do not know the true distribution from which our dataset is sampled, we cannot directly compute the true risk, given by

$$R(f) = \mathbb{E}_{(X,Y)\sim p}\left[(Y - f(X))^2\right],$$

of any model. Our best learning machine is the one that makes the empirical risk as small as possible. Mathematically this empirical risk is given by

$$R_n(f) = \frac{1}{n}\sum_{i=1}^{n}(y_i - f(x_i))^2.$$

In context the Empirical Risk Minimization is just taking the learning machine whose predictions are as close as possible (without overfitting) to the true values. The squared error loss is used here due to the fact that for this loss function larger mistakes are penalized more heavily than smaller ones, ensuring that the model focuses on avoiding big mispredictions. Additionally since the squared error loss is differentiable everywhere then the computations related to finding the best learning machine using this loss function become easier to handle.

Wheel 5: Algorithmic and Computational Optimization

Since the chosen function space inherently suggests linearity in the parameters, then some possibiliites for an algorithm include the `Ordinary Least Squares(OLS)` optimization, `Maximum Likelihood Estimation(MLE)`, and `Bayesian Learning`. In this case an assumption of the independence of the error terms and constant variance among these error terms in turn means that the `MLE` and `OLS` estimates will coincide.

In the case of Bayesian Linear Regression, it requires that prior distributions are placed on regression coefficients. I do not have enough domain knowledge that would justify valid priors, and using non-informative priors would essentially reduce the method to a more complex version of `OLS`. In that regard I proceed with the `OLS/MLE` algorithm. Theoretically, this choice of algorithm is sound since the estimates of the parameters from `OLS` are unbiased.

The table below gives a summary of the estimates of the parameters using the OLS algorithm.

Table 2: OLS estimates using all variables p-values of 0 mean the p-values are very small and not actually 0

Term	Estimate	Std. Error	t value	p value
(Intercept)	-26543.2793	6987.9921	-3.7984	0.0002
Date	6.9386	1.6077	4.3158	0.0000
Age	-0.9235	0.1570	-5.8817	0.0000
MRT_distance	0.0008	0.0010	0.7843	0.4334
Convenience	0.4317	0.2130	2.0271	0.0435
Latitude	313.0627	45.1529	6.9334	0.0000
Longitude	39.8424	49.8424	0.7994	0.4247
sqAge	0.0175	0.0038	4.5760	0.0000

Term	Estimate	Std. Error	t value	p value
logMRT	-6.2714	0.9349	-6.7080	0.0000
longlat	NA	NA	NA	NA

The significant predictors in the model; `Date`, `Age`, `Convenience`, `Latitude`, `sqAge` (the square of the `Age` variable), and `logMRT` (the log of the MRT_distance) all have p-values below 0.05, indicating they meaningfully influence house prices. Looking at the direction of their effects, older properties and higher `logMRT` values tend to lower the `Price`, while bought later, those with higher `Convenience`, larger `Latitude`, and the squared `Age` positively impact the `Price`. On the other hand, `MRT_distance`, `Longitude`, and `longlat` (interaction term between latitude and longitude) do not appear to contribute significantly, with `longlat` being dropped due to multicollinearity.

Wheel 6: Regularization and Refinement

Based on the estimates displayed in the table in the previous wheel, it is observed that some estimates are not contributing much to the model itself. In that regard I proceed by using the Akaike Information Criterion (AIC) for stepwise model selection. AIC provides a balance between model fit and complexity, penalizing unnecessary predictors while rewarding improved explanatory power. This approach helps refine the model, ensuring that only the most relevant features are retained before evaluating its generalization performance. After I use k-fold cross validation to assess the generalizability of the model. This ensures that the model selected by AIC performs well on unseen data and is not just overfitting. The reason for using k-fold and not `Leave One Out Cross Validation` (LOOCV) is to avoid a high computational complexity. Additionally I want to avoid a high variance in error estimates. 10-fold cross-validation provides a good balance between bias, variance, and computation efficiency, allowing us to reliably estimate the model's generalization performance on unseen data.

Table 3: Coefficients and Significance of the AIC-Selected Model (AIC = 2342.02). p-values of 0 mean the p-values are very small and not actually 0

Term	Estimate	Std. Error	t value	p value	Significance
(Intercept)	-21474.0501	3313.4459	-6.4809	0.0000	***
Date	6.9682	1.6040	4.3442	0.0000	***
Age	-0.9275	0.1565	-5.9254	0.0000	***
Convenience	0.4427	0.2118	2.0900	0.0374	*
Latitude	301.5125	40.9503	7.3629	0.0000	***
sqAge	0.0176	0.0038	4.5987	0.0000	***
logMRT	-5.8880	0.5938	-9.9160	0.0000	***

Based on the AIC stepwise selection, the model retained six predictors: `Date`, `Age`, `Convenience`, `Latitude`, `sqAge`, and `logMRT.` Indicating that per the AIC these variables are the ones that capture the variability in `Price` without increasing the model complexity with `Age` and `logMRT` decreasing price, while `Date`, `Convenience`, `Latitude`, and `sqAge` increase it. The AIC decreased from 2345.23 in the full model to 2342.02 in the selected model, indicating an improved balance between model fit and complexity. Predictors removed by the procedure, such as Longitude and MRT_distance, were deemed non-essential, resulting in a simpler and more interpretable model.

Table 4: CV Performance Metrics for Selected AIC Model

	Value
RMSE	7.775
Rsquared	0.683

	Value
MAE	5.347
RMSE SD	2.548
Rsquared SD	0.133
MAE SD	1.050

Based on 10-fold cross-validation, the AIC-selected model demonstrates average generalizability. The average RMSE is 7.78, indicating that predictions deviate from actual house prices by roughly this amount on average. The R^2 of 0.68 shows that about 68% of the variability in house prices is explained by the model. The relatively small standard deviations across the folds indicate that the model performs consistently, regardless of which subset of the data is used. This shows that the selected model is not only simple and easy to interpret but also reliably predicts house prices.

Wheel 7: Extrinsic Predictive Comparisons

After using AIC to choose a model and cross-validation to make sure it is stable, the next step is to assess how well the model predicts new, unseen data i.e. the test set. This helps us determine how accurate the model is and whether the patterns it picked up from the training data are actually reliable. We can assess how well the model generalizes by contrasting the predicted and actual house prices. Theoretically, this is just another method of ensuring that the model maintains low prediction errors when presented with new data and is able to work well when deployed.

Table 5: Performance of the AIC-Selected Model on the Test Set

Metric	Value
RMSE	6.778
MAE	4.842
R-squared	0.698

On the test set, the AIC-selected model performs quite well. The RMSE of 6.78 shows that predicted house prices typically differ from the actual prices by about 6.8 units, while the MAE of 4.84 indicates that the average error is slightly smaller. With an R^2 of 0.70, the model captures roughly 70% of the variation in house prices, which is good enough for this kind of data.

Wheel 8: Statistical Inference and Theoretical Justification

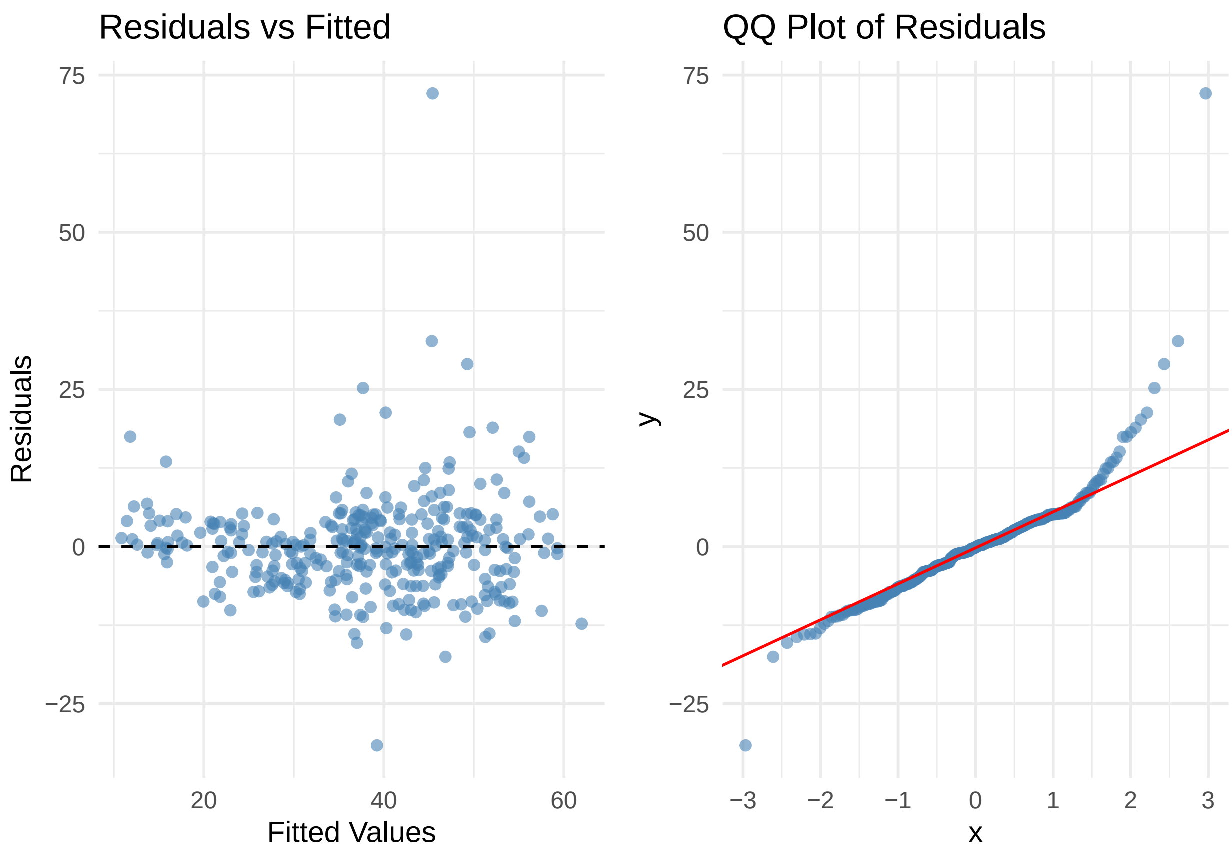
For this wheel, confidence intervals and p values were again used to test for the significance of each parameter estimate, qqplots and residual plots were used to check that the main theoretical assumptions on the error terms in Linear regression with the `OLS/MLE` algorithm that was chosen actually hold. In our regression problem we established that the OLS and MLE algorithms produce similar esimtates. For the OLS algorithm no distributional assumption is made on the error terms but these terms are assumed independent and with constant variance and 0 expectation. For the MLE we assume that the error terms are still independent but are normally distributed with constant variance and 0 expectation. The two plots chosen in this section allow us to check these assumptions on the residual terms. The plots and a table summarizing the significance of each estimate are given below:

Table 6: Coefficient Estimates with Confidence Intervals and Significance

Term	Estimate	Std. Error	t value	p value	CI Low	CI High	Significance
(Intercept)	-21474.0501	3313.4459	-6.4809	0.0000	-27992.5592	-14955.5409	***

Term	Estimate	Std. Error	t value	p value	CI Low	CI High	Significance
Date	6.9682	1.6040	4.3442	0.0000	3.8126	10.1239	***
Age	-0.9275	0.1565	-5.9254	0.0000	-1.2355	-0.6196	***
Convenience	0.4427	0.2118	2.0900	0.0374	0.0260	0.8595	*
Latitude	301.5125	40.9503	7.3629	0.0000	220.9513	382.0736	***
sqAge	0.0176	0.0038	4.5987	0.0000	0.0101	0.0251	***
logMRT	-5.8880	0.5938	-9.9160	0.0000	-7.0561	-4.7198	***

In general all predictors of our best selected model are significant which is indicated by both the low p-value and the nature of the confidence intervals.



- Residuals vs Fitted: The points are scattered fairly evenly around the horizontal line at 0, which suggests that the variance of residuals is roughly constant (no strong heteroscedasticity) and that the residual terms are independent of each other. There are a few noticeable outliers, but most residuals cluster around zero, indicating that the model is capturing the main trends in the data well.
- QQ Plot: Most of the points lie close to the red line, indicating that the residuals are approximately normally distributed. Some deviation at the extreme ends shows minor departures from normality, which is common in real-world datasets.

Our final selected model for predicting house prices in the Taiwan district is thus given as

$$\text{Price} = -21474.05 + 6.97\text{Date} - 0.93\text{Age} + 0.44\text{Convenience} + 301.51\text{Latitude} + 0.02\text{Age}^2 - 5.88\log\left(\text{MRT\_distance}\right)$$

Wheel 9: Deployment and Practical Scalability

Deployment is where the model is made available for use in the sense that it is used for its intended purpose by the people for whom it was made. In this `Taiwan House Price` context this means that this model is made available for use by stakeholders like buyers, sellers, and even policymakers. This is where the model's true generalizability is tested. For this project, the code and workflows were uploaded to a public GitHub repository which can be accessed here.

2. Practical Machine Learning - Digit Recognition

1. Data Preparation

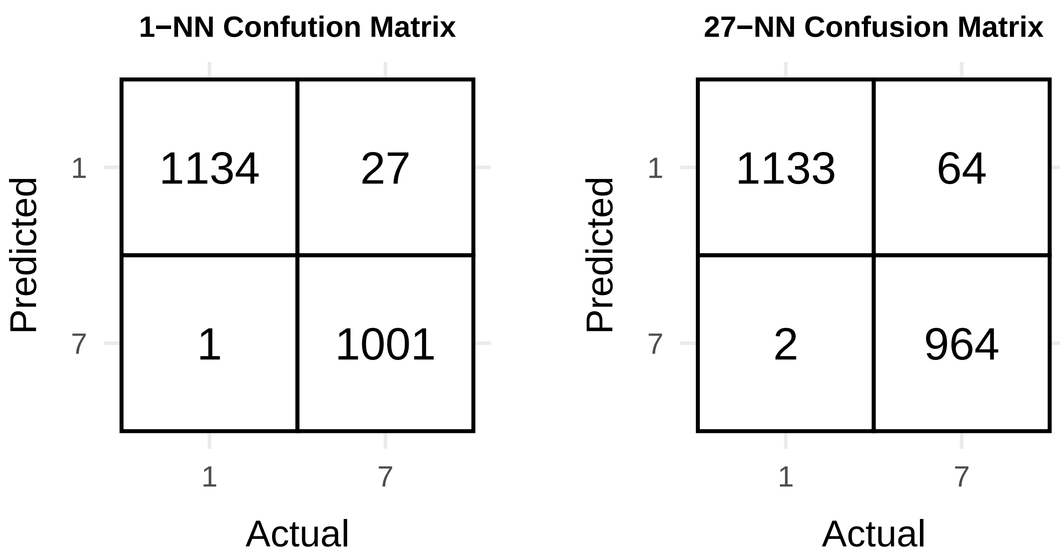
In this section the focus was to perform binary classification on a subset of the MNIST digit dataset. The dataset was prepared and a subset of the data, those whose labels are 1(positive class) and 7(negative class), was used for this classification task. Since the MNIST data already comes in a test set and a train set, those were the partitions that were used. Due to the large size of the training data, only 10% of the training data was used to train the model. In this task we classify the digits using the k-Nearest Neighbors (kNN) algorithm. The `MNIST` training and test labels were filtered for only the numbers 1 and 7. The main training set then was of size 13007 while the stratified sample contained 1300 samples. The number of features in the data set is 784. The choice of using a stratified sample for training was to make training faster, while maintaining class distribution occurring in the original training data.

2. Model Training

In this section we train two $k-$Nearest Neighbors classifiers based on the stratified sample of the training data. The choice of the kNN classifiers was due to that fact that these models are simple and can handle high dimensional data effectively.

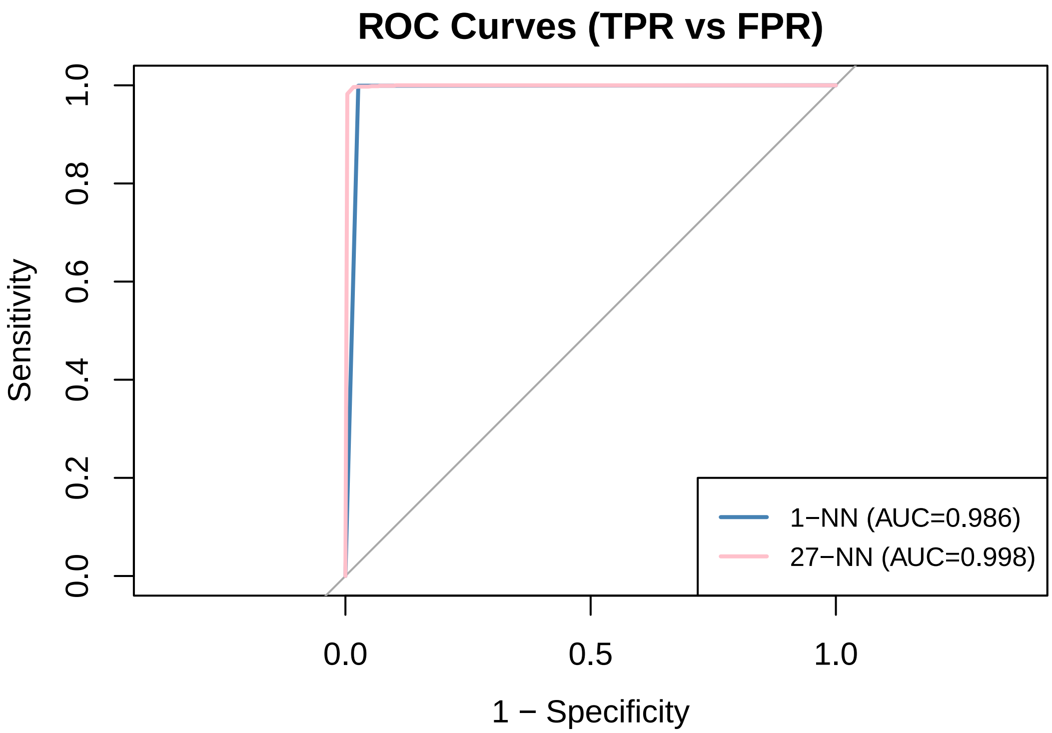
3. Performance Metrics

The accuracy and confusion matrix for each of these models is given below.



Based on the confusion matrices, the $1-$NN classifier seems to be performing better with 2135 images being correctly predicted while 28 are being wrongly predicted as opposed to the $27-$NN where 2097 images being correctly predicted while 66 are being wrongly predicted. The accuracy for the $1-$NN model is 98.7055 while the accuracy for the $27-$NN model is 96.9487 which is consistent with what is being displayed in the scatter plots.

4. ROC Curve Analysis

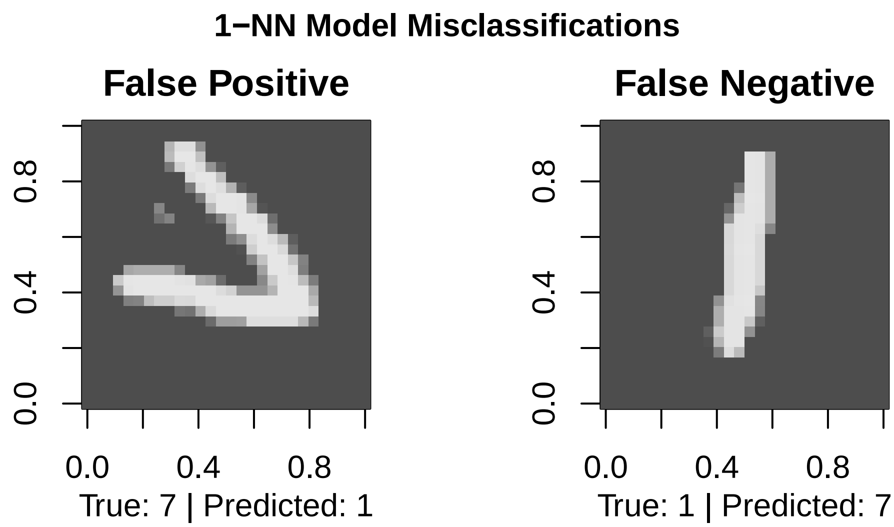


The ROC curve plots the `True Positive Rate`(Sensitivity) against the `False Positive Rate` (1-Specificity) for the two $k-$NN models that were trained. In particular both models perform well at correctly classifying images as `1` since the ROC curves for both models like well above the diagonal, indicating high sensitivity. The 27−NN model shows a slightly higher sensitivity at the same false positive rate as compared to the 1−NN model, suggesting that the former is better at correctly identifying class 1 images. This is consistent with the `Area Under the Curve`(AUC) values indicated in the legend.

While the confusion matrix shows that the 1−NN model correctly predicts more instances in each class than the 27−NN model, resulting in higher accuracy at the chosen threshold, the ROC curves tell a slightly different story. The AUC for the 27−NN model is higher than that of 1−NN, indicating that, across all possible thresholds, the 27−NN model has a slightly better ability to discriminate between the classes. This highlights an important distinction: accuracy from the confusion matrix evaluates performance at a single threshold, whereas AUC summarizes the model's discriminative ability across thresholds.

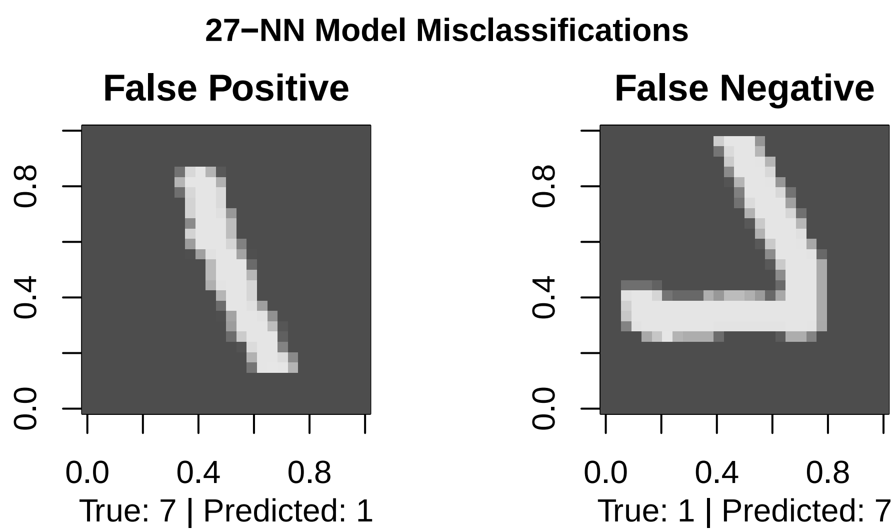
5. Error Visualization

In this section the plot of some false positives and false negatives, as observed from the model predictions are displayed.



For the 1−NN model we see some examples of false positives and false negatives. In particular we see that the model is predicting the image on the left as a 1 whereas the true value is obviously 7. The model may have made this mistake particularly because of the orientation of the image. In a likewise manner for the false negative where the true label is 1 while the model is predicting a 7. This could be due to the fact that the image looks cropped at the top.

For the example of the false positive for 27−NN model, the reason for this prediction could be in the fact the image looks cropped at the bottom, whereas for the false negative, the prediction could be due to the orientation as well as the cropped nature towards the top of the image. The images are given below.



3. Discovering the concept of ultra high dimensionality

For this section 6 different datasets were used. For each dataset, a brief overview of its dimensionality, sample size, and homogeneity is provided, considering both type-homogeneity (consistency of variable types) and scale homogeneity (consistency of measurement scales). The n/p ratio (number of observations over the number of features) is also computed. Finally, graphical summaries of a subset of variables are provided, and correlations and multicollinearity are examined where applicable, with the aim of understanding the structure and properties of each dataset.

The general outline for this section is as follows:

1. Download the dataset and open it. In this section the dataset is loaded and displayed to provide some understanding of its contents and structure.
2. Provide a succinct description of the dimensionality of the data, sample size and homogeneity. In this section some comments are made on the dimensionality of the data, its sample size the weather the features are homogeneous and if the scales are homogeneous as well.
3. Commenting on the quintessentially important index κ := n/p. This section gives an insight into the nature of the dimensionality of the data; whether the data is high dimensional or not.
4. Generating basic graphical summaries Some basic graphical summaries like the distribution of the features are given for only 9 of the variables in each dataset.
5. Discussions on correlations and multicollinearity Some comments are made on multicollinearity and correlations between these 9 variables.

DNA Data

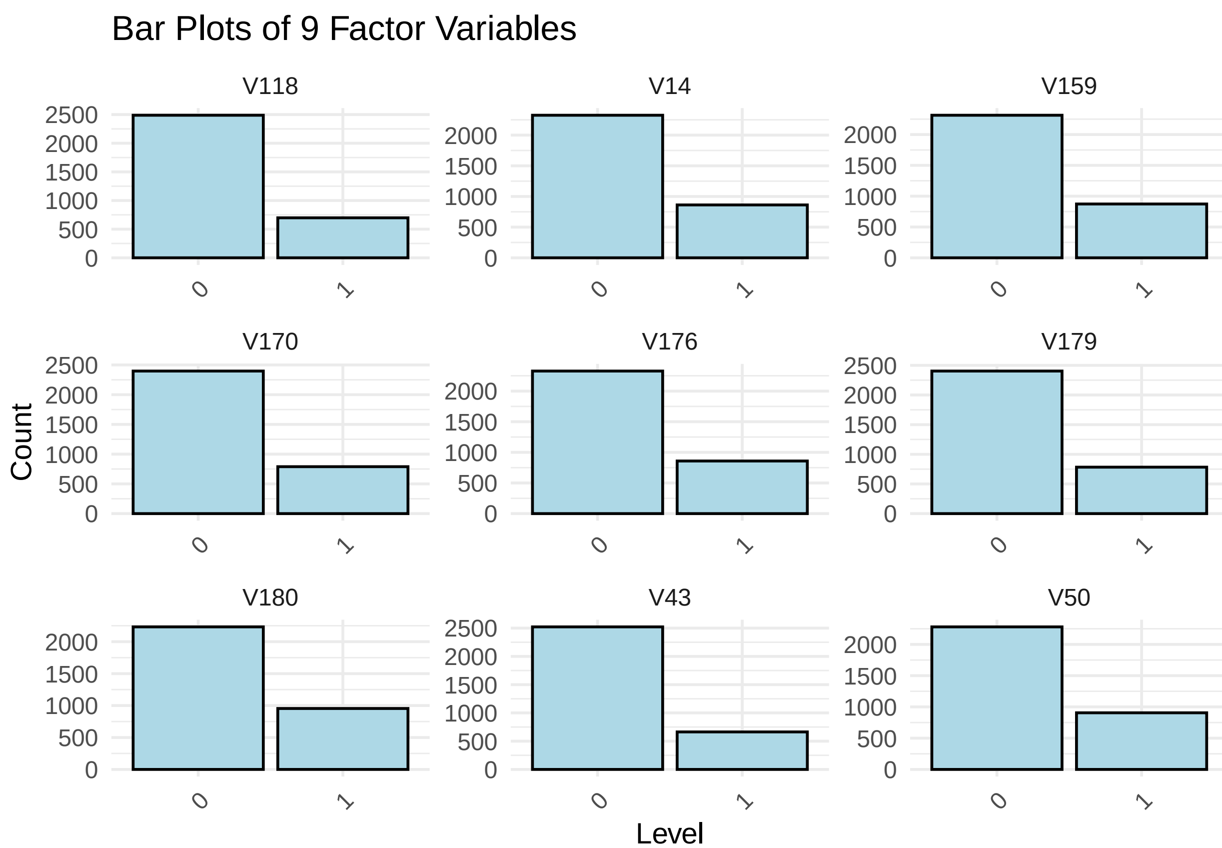
1. Below is a table of a few of the rows in the dataset.

V159	V179	V14	V170	V50	V118	V43	V180	V176
0	0	0	1	1	0	0	0	0
0	1	0	1	0	0	0	0	0
0	0	0	1	0	0	0	1	0

2. This dataset has 3186 observations and 180 features. The features are of 1 different data types (Categorical) among the features. Even though the features look like integer values, they are essentially categorical values that have been encoded, possibly with the usual `label encoding`. In that regard it does not make sense to talk about scale homogeneity.

3. The index κ for this dataset is 17.7 this indicates that the number of samples are more than the observed features (an overdetermined system). This suggests that machine learning estimators are likely to be stable, and the risk of overfitting is relatively low. The large κ value implies there is sufficient information in the data to reliably estimate model parameters.

4. In this case a high class imbalance is observed for all the 9 sampled categories, indicating that some feature categories dominate the others. In such a situation this can reduce the model's ability to learn from minority categories and limit the contribution of those features to predictive performance.



5. All predictors here are categorical, so Pearson correlation and traditional multicollinearity do not apply. Although categorical variables may still be associated (e.g., via Cramér's V), such associations do not introduce the $X^T X$-related instability seen in numerical regression models. Since we are performing multi-class classification, this is not a practical concern for our analysis.

Breast Cancer Data

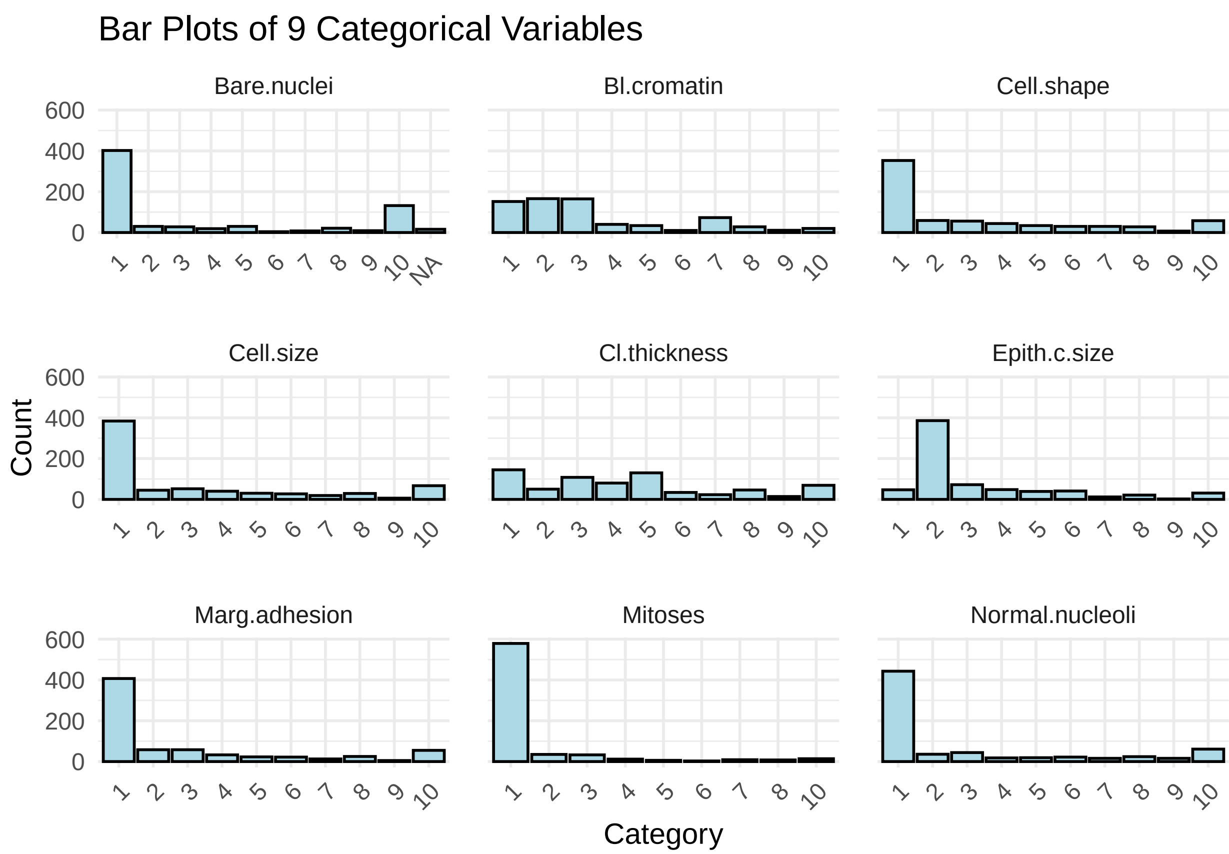
1. Below is a table of a few of the rows in the dataset.

Cell.size	Class	Cl.thickness	Epith.c.size	Mitoses	Marg.adhesion	Cell.shape	Normal.nucleoli	Bl.cromatin
1	benign	5	2	1	1	1	1	3
4	benign	5	7	1	5	4	2	3
1	benign	3	2	1	1	1	1	3

2. This dataset has 699 observations and 9 features. The features together with the response are all categorical (encoded to numeric). The variables are, therefore, type homogenoeus. Since the features are categorical, it does not make sense to talk about scale homogeneity as the encoded values are really just discrete values. So the notion of scale homogeneity does not apply in the previous sense, although different variables have different numbers of classes.

3. The index κ for this dataset is 61.545 this indicates that the number of samples are more than the observed features (an overdetermined system). This suggests that machine learning estimators are likely to be stable, and the risk of overfitting is relatively low. The large κ value implies there is sufficient information in the data to reliably estimate model parameters.

4. These bar plots show that almost all nine predictors are heavily concentrated in the lowest category, with very few observations falling into higher abnormality levels. This strong skew suggests that most patients present with mild cellular features, while severe morphological changes are relatively rare in the dataset. Such imbalanced category distributions may influence how models learn patterns, since the majority of information comes from the lower-grade classes.



5. Although the response is binary, all predictors are categorical. This means multicollinearity diagnostics like Pearson's correlation are not appropriate because the model uses dummy-coded factors rather than continuous numeric predictors. Categorical predictors may still show associations (e.g., via chi-square or Cramér's V), but these do not produce the linear-dependence (singularity of $X^T X$ matrix) issues seen in numeric regression models. Therefore, multicollinearity is not a practical concern here.

Spam Data

1. Below is a table of a few of the rows in the dataset.

num650	project	internet	re	capitalLong	business	order	free	font
0	0	0.00	0.00	61	0.00	0.00	0.32	0
0	0	0.07	0.00	101	0.07	0.00	0.14	0
0	0	0.12	0.06	485	0.06	0.64	0.06	0

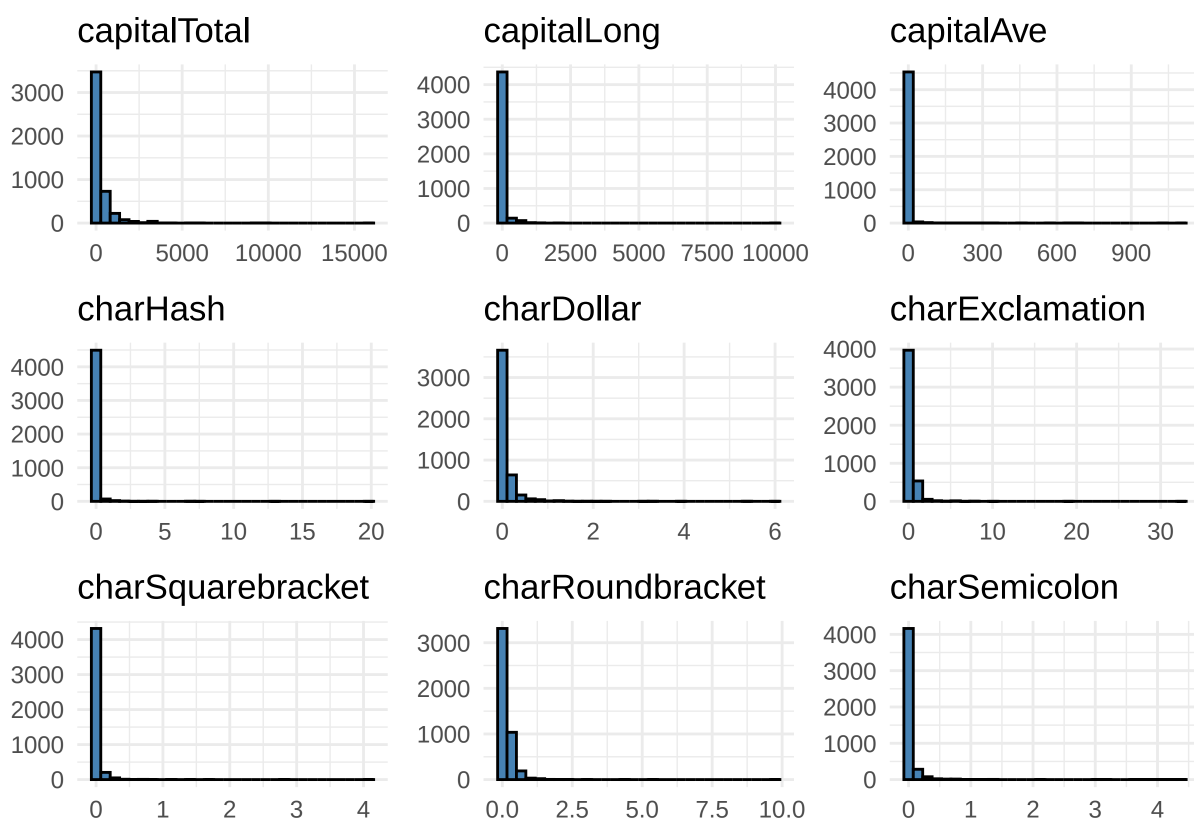
2. This dataset has 4601 observations and 57 features. The features are all numeric while the response is categorical, making this a classification problem. A table of some of the numeric features are presented below indicating a heterogeneous scale for the numeric features.

Table 10: Numeric features with scale summary

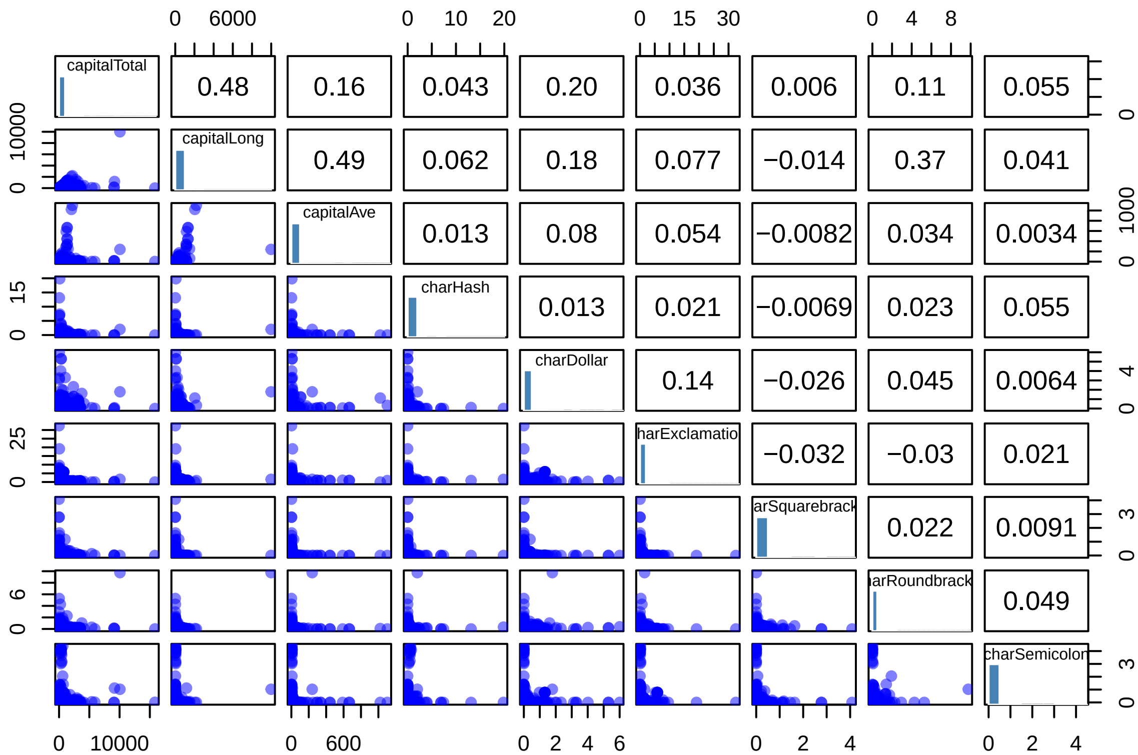
Feature	Min	Max
num3d	0	42.81
meeting	0	14.28
telnet	0	12.50
labs	0	5.88
lab	0	14.28

3. The index κ for this dataset is 80.719 this indicates that the number of samples are more than the observed features (an overdetermined system). This suggests that machine learning estimators are likely to be stable, and the risk of overfitting is relatively low. The large κ value implies there is sufficient information in the data to reliably estimate model parameters.

4. These histograms show that some features are extremely right-skewed, with most emails having values clustered very close to zero. A small number of emails contain unusually high counts of characters or capital letters, creating long tails in several distributions. Overall, this suggests that while most messages use these features, a few outliers may strongly influence how a classifier identifies spam-like behaviour.



5. The pairwise correlations among the nine features are all quite small, with the strongest relationship around 0.48 between capitalTotal and capitalLong. This indicates some mild association but nothing close to the levels (e.g., >0.8) that raise multicollinearity concerns. The scatterplots confirm this: the relationships are weak, scattered, and mostly nonlinear, suggesting that each feature contributes different information about character usage in emails. Overall, the low correlations mean collinearity is minimal, and multicollinearity is not an issue. These predictors can coexist comfortably in a model without causing instability or inflated variance.



Leukemia Data

1. Below is a table of a few of the rows in the dataset.

x.2462	x.2510	x.2226	x.525	x.194
-1.4140952	0.9626709	0.0920567	-0.4909099	0.3759408
-0.8169876	0.7307726	0.7381219	-0.6230732	1.2540752
-0.3412622	0.8111270	-0.0531827	0.3076012	0.3591718

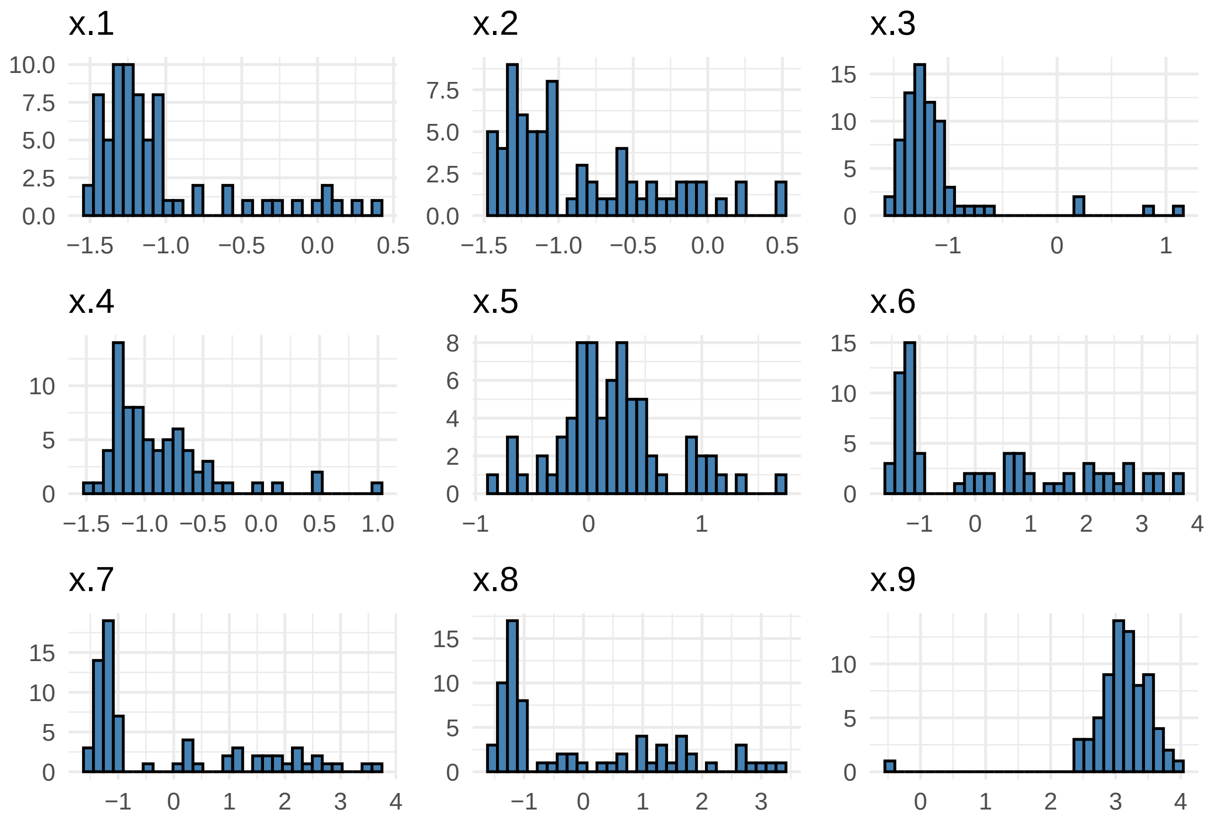
2. This dataset has 72 observations and 3571 features. There is only 1 different data type (numeric) among the features. Below is a table of some of the numeric features with their scales, indicating a heterogeneous scale.

Table 12: Numeric features with scale summary

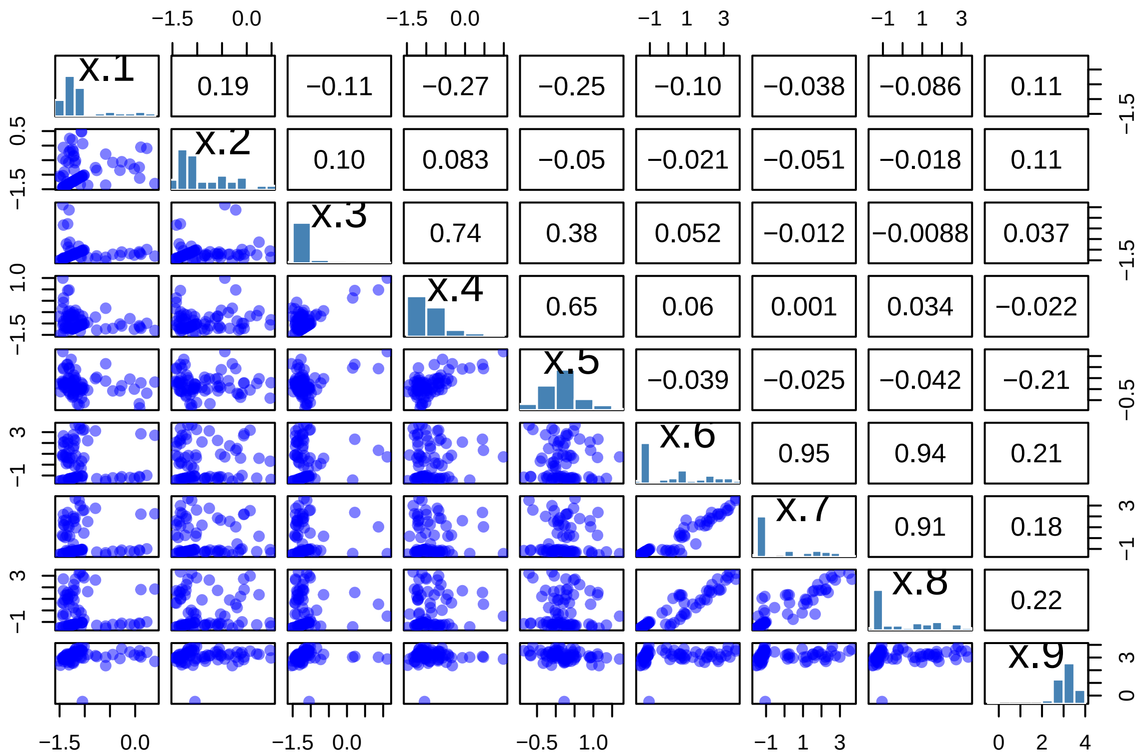
Feature	Min	Max
x.2985	-1.4243923	1.342014
x.1841	-1.2545862	1.350872
x.1141	-0.2267222	2.225467
x.3370	-1.4816277	2.793430
x.1252	-1.2463837	0.774674

3. The index κ for this dataset is 0.02 this indicates that the number of samples are way less than the observed features (an underdetermined system). This suggests that machine learning estimators are likely to be unstable and prone to overfitting, unless strong regularisation or dimensionality reduction is applied. The small κ value implies that the data contain limited information for reliably estimating model parameters.

4. These histograms show that most of the variables are right-skewed, with many observations clustered on the left and a long tail extending to the right. A few variables (like x.5 and x.9) appear more balanced or centered, while others have clear outliers stretching the distribution. Overall, the features vary in shape, but many display skewness that may benefit from transformation depending on the modelling approach.



5. The correlations here are mostly very small, which means the variables are only weakly related to one another. A few pairs such as X.3–X.4 and X.6–X.7 show moderate positive relationships, but nothing approaching the high correlations (e.g., >0.8) that typically cause multicollinearity issues. Overall, the scatterplots look diffuse rather than tightly linear, so collinearity is minimal and multicollinearity is not a practical concern for modelling with these predictors.



Prostate Data

1. Below is a table of a few of the rows in the dataset.

	X214065_s_at	X217522_at	X221420_at	X215861_at	X202735_at
PG13	-0.3241601	-0.3375143	-0.3563790	-0.3141736	-0.1278737
PG15	-0.2696562	-0.2841149	-0.2764088	-0.2298867	-0.1214101
PG37	-0.2503474	-0.2518213	-0.2638858	-0.2083398	-0.1121377

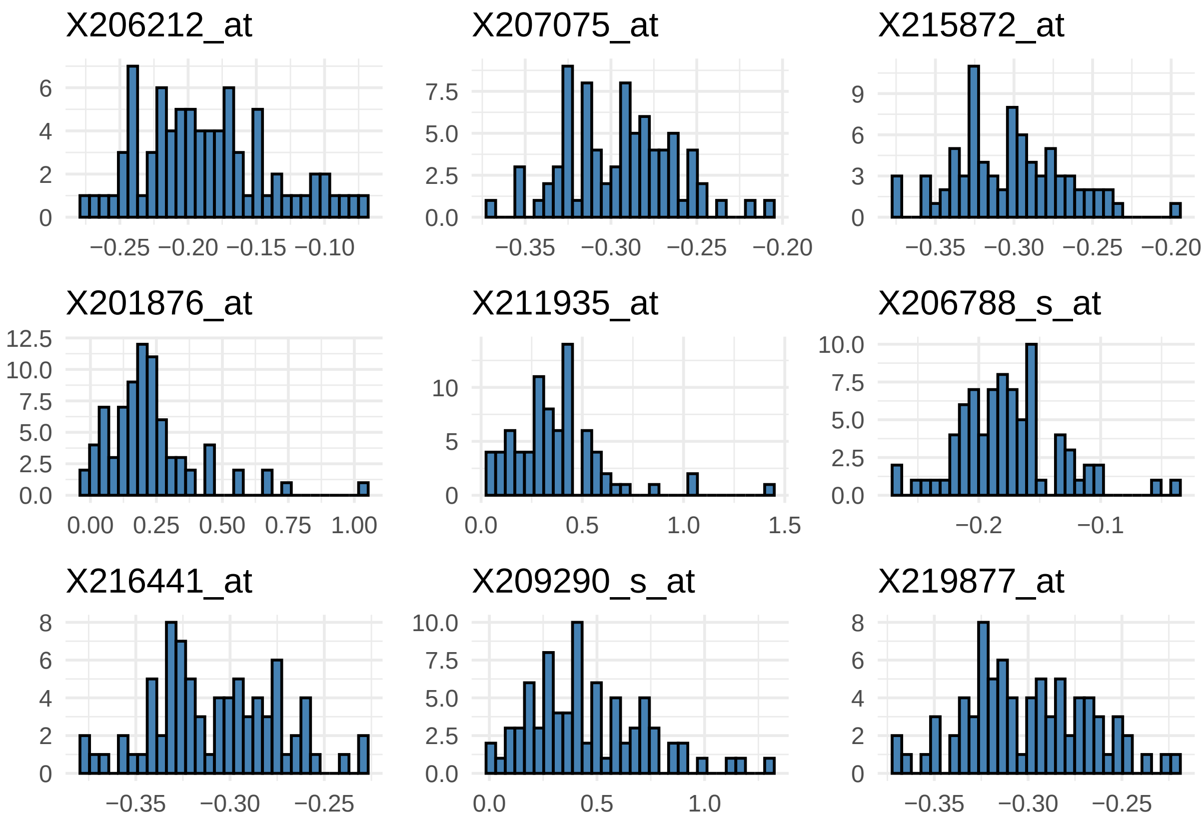
2. This dataset has 79 observations and 500 features. All features are numeric, (type homogeneity). A table with the scales of some of these numeric features are given below. These scales indicate scale heterogeneity.

Table 14: Numeric features with scale summary

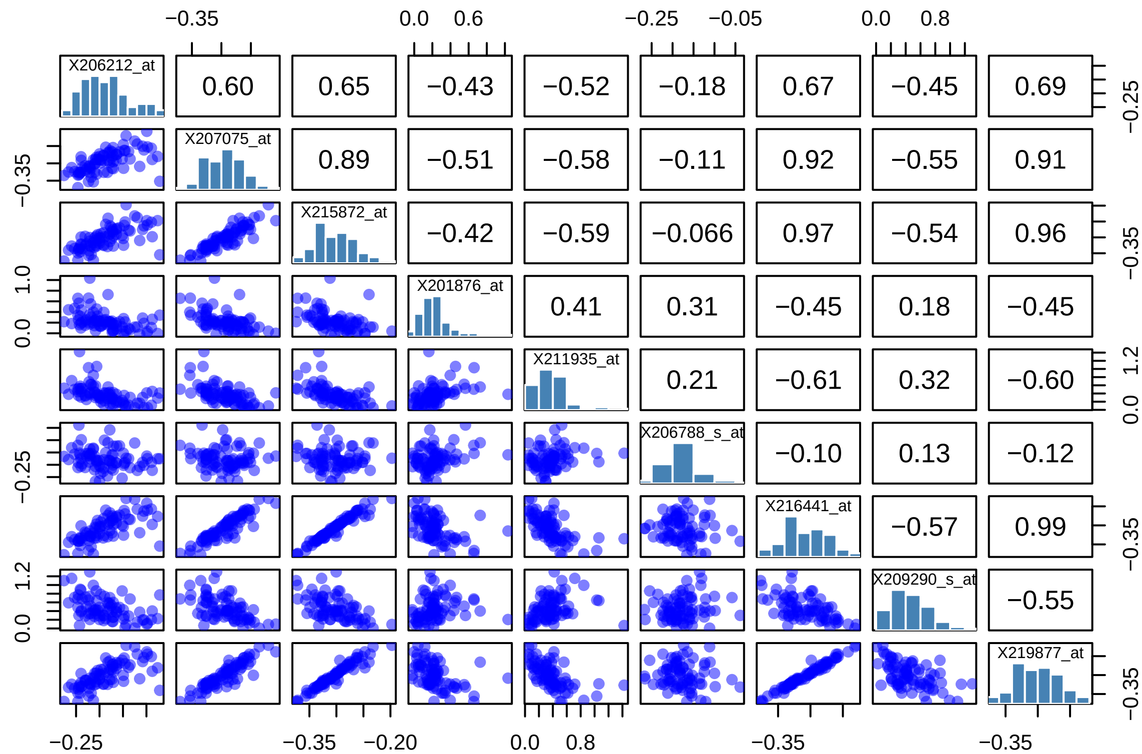
Feature	Min	Max
X206465_at	-0.3627913	-0.2106258
X213166_x_at	0.0439634	1.6443740
X215738_at	-0.3446781	-0.1440000
X213447_at	-0.1189823	0.4761519
X220488_s_at	-0.2557114	-0.0680252

3. The index κ for this dataset is 0.158 this indicates that the number of samples are less than the observed features (an underdetermined system). This suggests that machine learning estimators are likely to be unstable and prone to overfitting, unless strong regularisation or dimensionality reduction is applied. The small κ value implies that the data contain limited information for reliably estimating model parameters.

4. These histograms show that some gene expression features have thier own distinct distribution, with most values clustering within a narrow range and only a few observations stretching into the tails. Some genes appear more symmetrically distributed, while others display noticeable right-skewness, suggesting occasional higher expression levels. Overall, the plots highlight that the genes vary in both spread and shape, which may help explain differences in how strongly each one contributes to downstream modelling.



5. These pairwise plots show that several gene-expression features are strongly correlated, with many correlation values above 0.9. The scatterplots for these pairs form very tight linear patterns, confirming that the variables move almost identically. Such high correlations indicate serious collinearity, meaning these genes carry nearly the same information. When used together in a regression or similar model, they would create multicollinearity, leading to unstable coefficient estimates and difficulty separating each gene's individual effect.



Colon Data

1. Below is a table of a few of the rows in the dataset.

X414	X462	X178	X525	X194
0.9183690	0.6105782	1.048785	0.9845771	0.9797156
0.2139728	-0.6466094	1.524639	0.9453600	0.6979945
1.0127060	0.3355917	0.593305	1.2700206	1.6433465

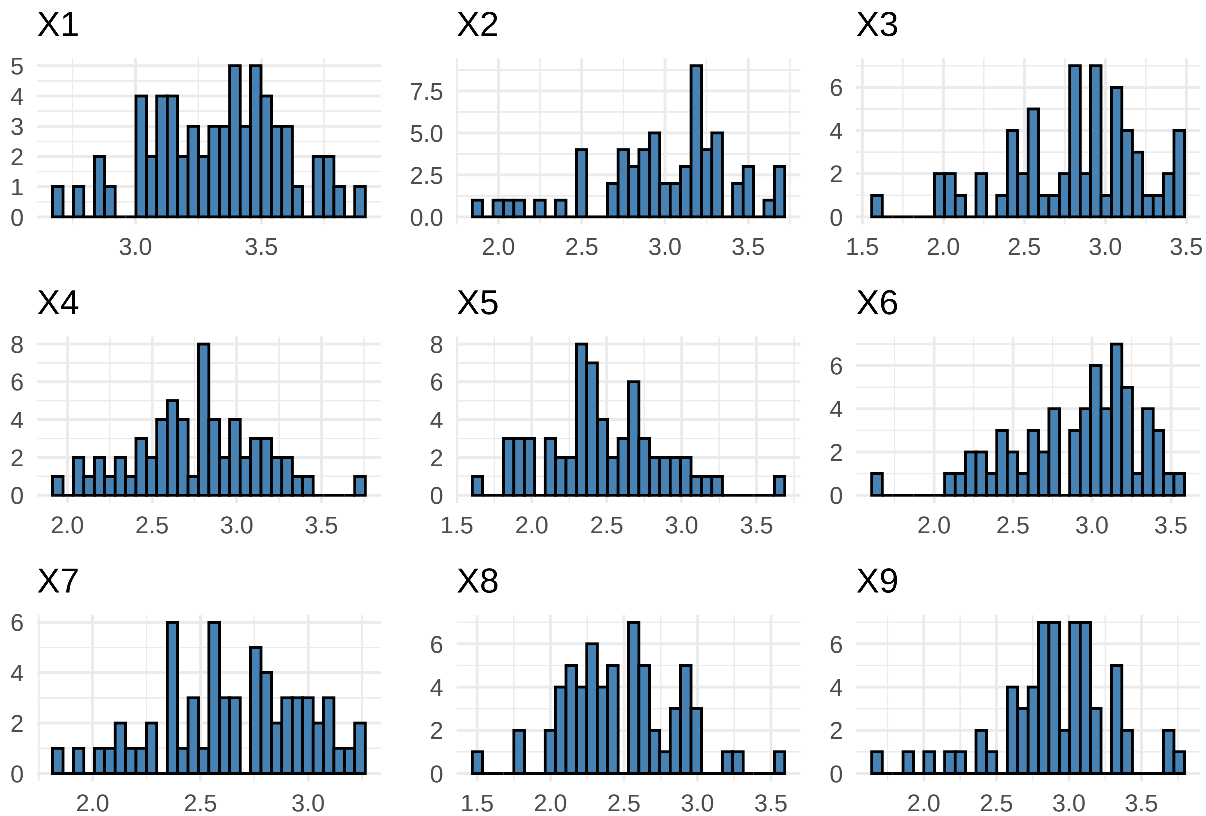
2. This dataset has 62 observations and 2000 features. The features are all numeric. The table below shows some of the features and their scales, indicating scale heterogeneity.

Table 16: Numeric features with scale summary

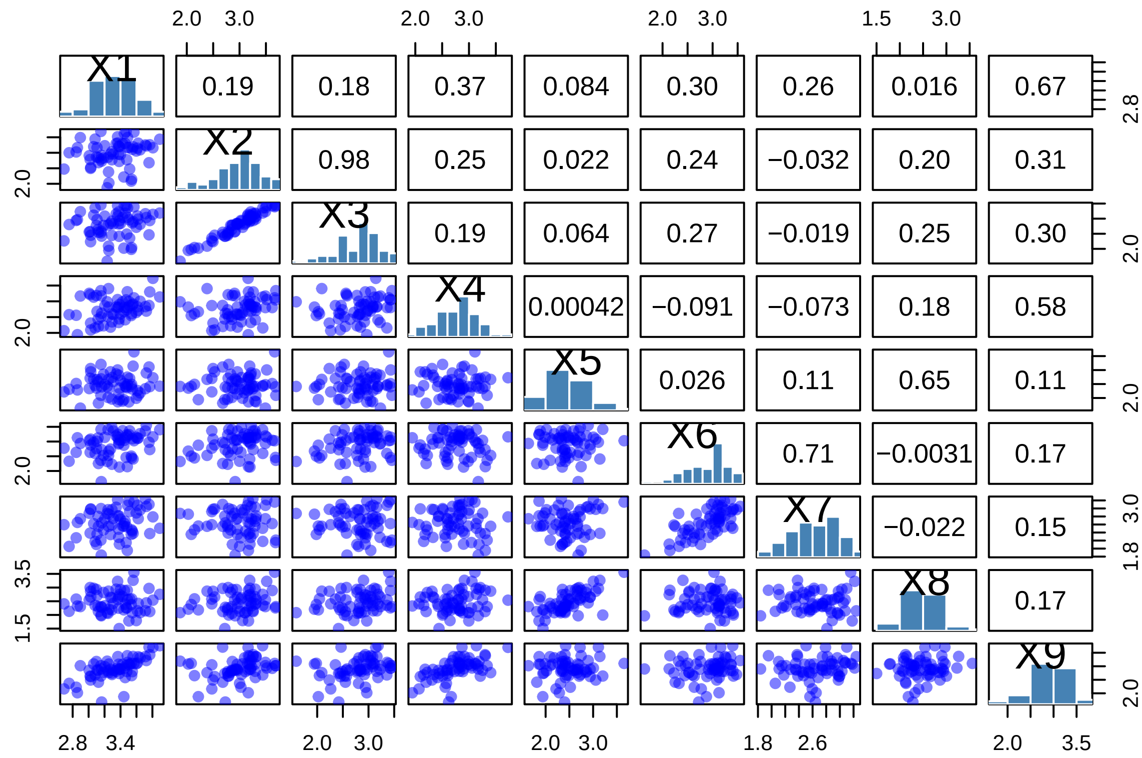
Feature	Min	Max
X937	-1.421323	0.6851193
X1841	-3.256031	0.1303733
X1141	-1.511940	0.9222931
X1322	-1.887731	-0.0958678
X1252	-1.620780	0.3181653

3. The index κ for this dataset is 0.031 this indicates that the number of samples are less than the observed features (an underdetermined system). This suggests that machine learning estimators are likely to be unstable and prone to overfitting, unless strong regularisation or dimensionality reduction is applied. The small κ value implies that the data contain limited information for reliably estimating model parameters.

4. These histograms show that all nine variables follow roughly bell-shaped distributions, though each with slightly different centers and spreads. Some variables, like X5 and X7, appear more tightly concentrated, while others such as X2 and X6 are more spread out across their ranges. Overall, the features look reasonably normal, with no extreme skewness or heavy outliers, suggesting they're suitable for modelling without major transformation.



5. This plot shows that most variable pairs have very low correlations, meaning the predictors are largely independent of one another. Only a couple of relationships stand out, like X1 to X2 and X6 to X7 but these correlations are moderate rather than extreme. The relation between X2 and X3 standout because of the nature of the scatter plot as well as the high correlation coefficient, indicating some multicollinearity. Overall, most variables do not exhibit the strong linear relationships that cause multicollinearity, even though some of them do. These variables that do exhibit a high multicollinearity could destabilize the model.



4. Investigating the Bayes Risk R^* in Regression

Let X and Y be two continuous random variables with joint probability density function

$$p(x,y) = \frac{1}{2\pi\sqrt{2\pi\frac{9}{\pi^2}}} \exp\left\{-\frac{\pi^2}{18}\left[y - \frac{\pi}{2}x - \frac{3\pi}{4}\cos\left(\frac{\pi}{2}(1+x)\right)\right]^2\right\}.$$

It can be shown that the conditional density of Y given X is given by

$$p_1(y \mid x) = \frac{1}{\sqrt{2\pi\frac{9}{\pi^2}}} \exp\left\{-\frac{\pi^2}{18}\left[y - \frac{\pi}{2}x - \frac{3\pi}{4}\cos\left(\frac{\pi}{2}(1+x)\right)\right]^2\right\}, \qquad -\infty < x < \infty,$$

while the marginal density of X is given by

$$p_2(x) = \frac{1}{2\pi}, \qquad 0 \le x < 2\pi.$$

1. Find and write down the expression for $\mathbb{E}[Y \mid X]$

Notice that the conditional distribution of Y given X can be rewritten to get:

$$\frac{1}{\sqrt{2\pi\frac{9}{\pi^2}}} \exp\left\{-\frac{\pi^2}{18}\left[y - \frac{\pi}{2}x - \frac{3\pi}{4}\cos\left(\frac{\pi}{2}(1+x)\right)\right]^2\right\} = \frac{1}{\frac{3}{\pi}\sqrt{2\pi}} \exp\left\{\frac{-\left(y - \left(\frac{\pi x}{2} + \frac{3\pi\cos\frac{\pi(1+x)}{2}}{4}\right)\right)^2}{2\left(\frac{9}{\pi^2}\right)}\right\}$$

which is exactly the normal distribution with mean $\left(\frac{\pi x}{2} + \frac{3\pi \cos\frac{\pi(1+x)}{2}}{4}\right)$ and variance $\frac{9}{\pi^2}$

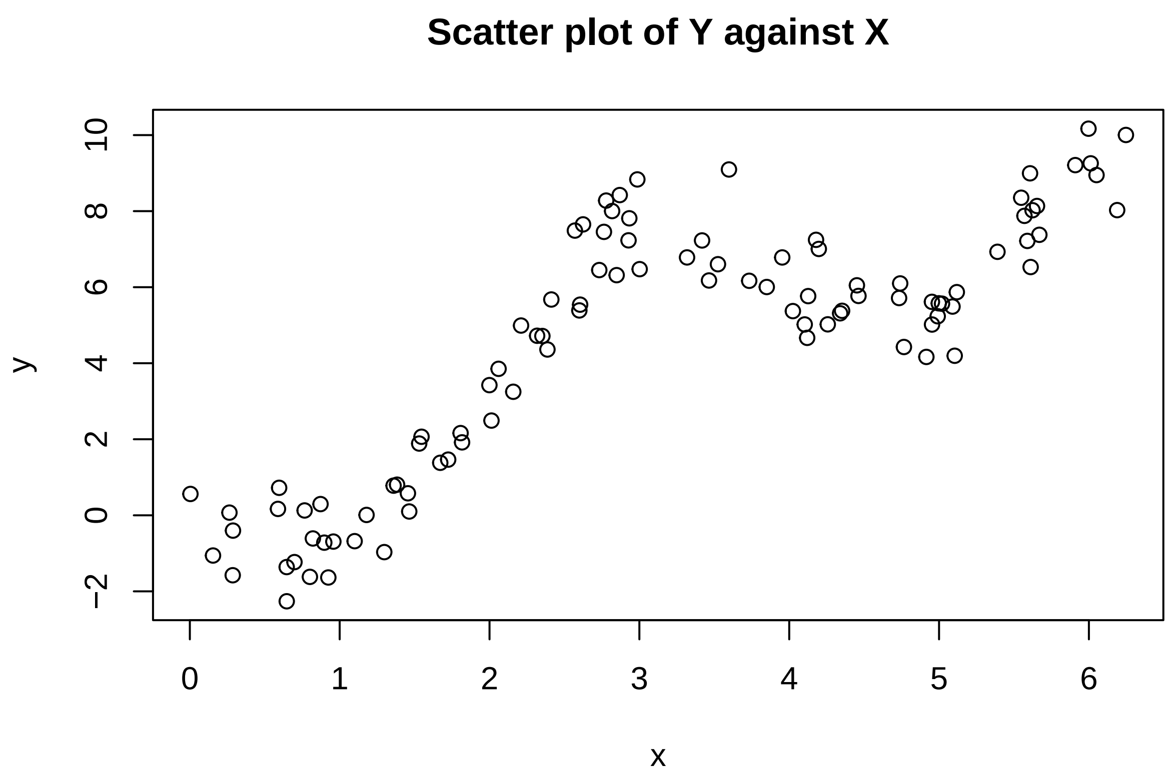
2. Generate an iid sample

Below is a table of the first 10 (x, y) pairs that were randomly generated.

Table 17: First 10 (x, y) pairs

	x	y
1	1.8069028	2.162056
2	4.9530672	5.017812
3	2.5696778	7.488250
4	5.5481620	8.352669
5	5.9091302	9.210446
6	0.2862399	-1.576140
7	3.3181846	6.782776
8	5.6072342	8.994681
9	3.4647684	6.176293
10	2.8689950	8.422993

3. Draw the scatter plot of the iid sample



The scatter plot shows a clear curved pattern, where Y increases and decreases smoothly as X varies. This aligns with the theoretical conditional mean, a sinusoidal function rather than a straight line. Although the individual points are scattered because of noise, the overall trend is visibly nonlinear but some parts of the plot look linear (from x=1 to x=3). This indicates that a linear combination of possibly linear and sinusoidal functions will accurately capture the trend in this data.

Question 4: Regression Analysis and Bayes Risk

Consider using regression analysis learning machines to learn the true function underlying the generated dataset. The loss function to be used is

$$\ell(Y, f(X)) = (Y - f(X))^2$$

with the corresponding functional risk

$$R(f) = \mathbb{E}[\ell(Y, f(X))] = \int_{\mathcal{X}\times\mathcal{Y}} \ell(y, f(x)) p_{XY}(x, y) dx dy$$

a. We want to find the expression of

$$f^*(X) = \text{argmin}_f\, \mathbb{E}[\ell(Y, f(X))]$$

but notice that this is given by

$$\underset{f}{\text{argmin}} \mathbb{E}[\ell(Y, f(X))] = \underset{f}{\text{argmin}} \mathbb{E}[\ell(Y, f(X))] = \underset{f}{\text{argmin}} \int_{\mathcal{X}\times\mathcal{Y}} \ell(y, f(x)) p_{XY}(x, y) dx dy$$

which becomes

$$\underset{f}{\text{argmin}} \int_{\mathcal{X}\times\mathcal{Y}} \ell(y, f(x)) p_{XY}(x, y) dx dy = \arg\min_{f} \int_{\mathcal{X}} \int_{\mathcal{Y}} (y - f(x))^2 p_{XY}(x, y)\, dy\, dx.$$

By differentiating with respect to x we get the above becomes

$$\arg\min_{a\in\mathbb{R}} \int_{\mathcal{Y}} (y - a)^2 p_{Y|X}(y \mid x)\, dy,$$

which implies that $a = \mathbb{E}[Y \mid X = x]$. Hence

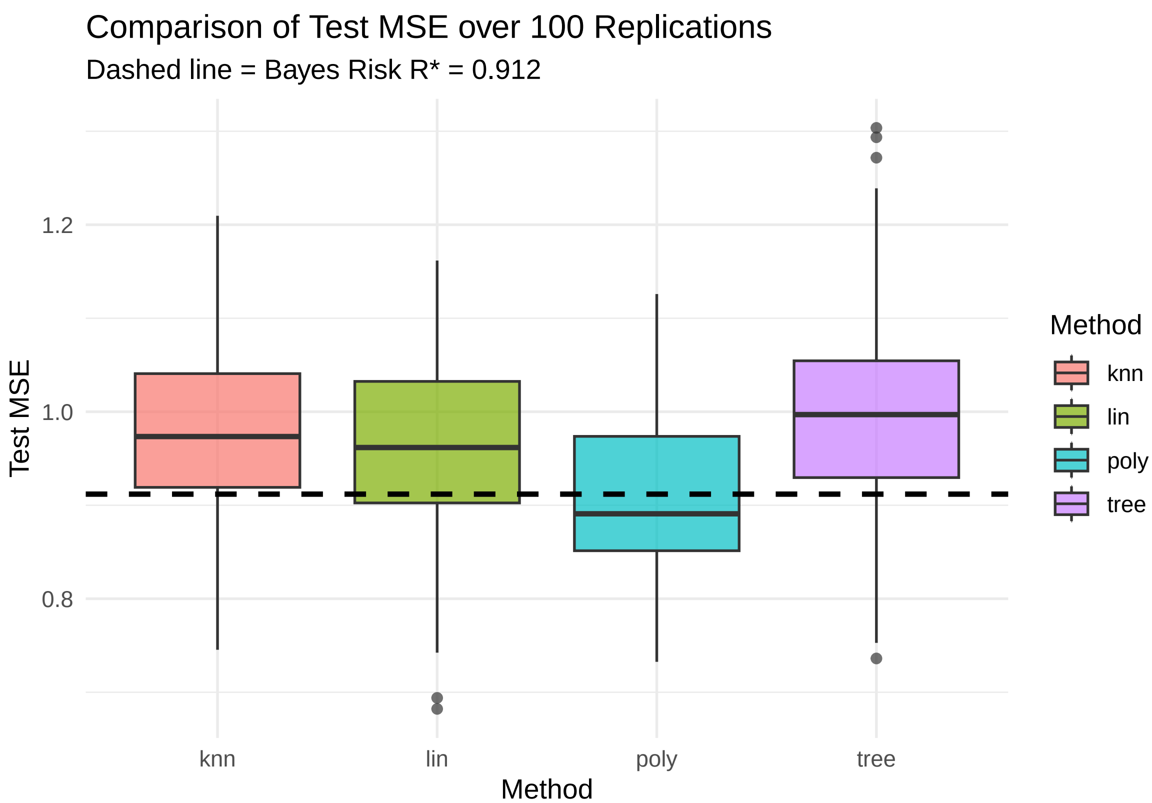
$$f^*(x) = \mathbb{E}[Y \mid X = x]$$

and for our model

$$f^*(x) = \frac{\pi}{2}x + \frac{3\pi}{4}\cos\Big(\tfrac{\pi}{2}(1 + x)\Big).$$

b. The Bayes risk equals the expected conditional variance which from the previous question we see is given as:

$$R^* = \mathbb{E}[\text{Var}(Y \mid X)] = \frac{9}{\pi^2}.$$



c.

This simulation shows significant differences in performance among the tested regression models. Our benchmark is the **Bayes Risk** (R^*), which is the absolute theoretical minimum achievable test error for this problem, calculated as $R^* \approx 0.912$.